

Maintenance and Tuning Guide

Safe changes for the offline Windows 11 Matrix deployment

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Item	Value
Homeserver URL	Environment-specific (POC: http://192.168.0.113)
Matrix identity	axon.home.arpa
Client networks	Flat LAN or routed/NATed private subnets
Host exposure	TCP 80 only

Axon provides text messaging through stock Element or Element X. Internet, cellular data, federation, media, calling, and internet push notifications are outside the initial proof of concept.

1. Routine operations

```
.\scripts\Test-Axon.ps1 -BindIp 192.168.0.113  
# Replace the example IP with the selected Windows address
```

Host-only Axon Control

```
http://127.0.0.1:8780
```

- Sign in with a Matrix server-administrator account; credentials are not stored.
- Create individual or 1-200 account batches with stock passwords and standard/admin roles.
- Download the issued-account CSV, distribute it privately, and delete it when no longer needed.
- Reset passwords, promote/demote, lock/unlock, and review honest last-seen activity categories.
- Inspect container health and CPU/memory use; start, stop, or restart a service or the full stack.
- Search rooms, create private encrypted rooms, inspect membership, and add or remove local users.
- Read bounded PostgreSQL, Synapse, and nginx logs.
- The nginx gateway blocks external Synapse admin paths; ports 8008 and 8780 remain loopback-only.

Room administration

- Axon-created rooms are private, E2EE-enabled, and non-federated.
- When changing membership in a user-created room, Axon visibly joins the signed-in administrator and grants room-level control. It does not silently impersonate another user.
- Delete and purge requires the exact room name, removes local members, blocks rejoining, and starts an asynchronous Synapse purge.
- Activity categories use last-seen timestamps. Presence is disabled, so they are not exact online/offline indicators.

Use whole-stack controls for routine work. Stopping PostgreSQL interrupts Synapse, stopping Synapse interrupts messaging, and stopping nginx removes client ingress. Pause all services preserves configuration, users, rooms, and volumes.

Protected state

%ProgramData%\Axon contains .env, homeserver.yaml, nginx configuration, secrets, and installer state. Docker volumes contain PostgreSQL and Synapse data. Back up before editing and never share these files.

2. Message retention

- Axon defaults to a 48-hour maximum lifetime with periodic purge.
- E2EE prevents Synapse from reading plaintext, but encrypted events and routing metadata are temporarily stored so offline recipients can synchronize.
- A 48-hour policy is not a guaranteed exact deletion timestamp; purge jobs run on an interval.
- Changing retention affects future server behavior and must be tested with disposable accounts before operational use.

```
Copy-Item $env:ProgramData\Axon\runtime\synapse\homeserver.yaml ` $env:ProgramData\Axon\runtime\synapse\homeserver.yaml.bak  
# Edit retention values, restart Synapse in Axon Control, then:  
.\scripts\Test-Axon.ps1 -BindIp 192.168.0.113
```

3. Resource tuning

- For the 200-user messaging POC, 4 or more modern CPU cores, 16 GiB RAM, and SSD/NVMe storage are recommended. The 32 GiB Razer provides useful headroom.
- Docker Desktop with WSL 2 shares Windows resources dynamically. Do not impose low WSL limits before collecting real usage data.
- Low RAM or disk produces an advisory warning during normal installation rather than blocking it. Record the warning and reduce the expected capacity.
- Keep 100 GiB free as the operational target and monitor any host below 20 GiB closely during image extraction and soak testing.
- Use Docker Desktop statistics and Windows Task Manager during a soak test; tune only after observing sustained pressure.

Safe change sequence

Step	Action
1	Copy the exact configuration file and record current container health
2	Make one change; never edit secrets or immutable image references casually
3	Restart only the affected service and run Test-Axon.ps1
4	Exercise two-client messaging and offline/reconnect delivery
5	Restore the backup immediately if health or messaging regresses

Release-level changes

Treat a Matrix identity-domain change, TLS/private CA, media, calling, federation, image upgrade, or nginx exposure as a planned release. Address/subnet changes require coordinated NIC, Compose, firewall, route/forward, and client-URL updates.

4. Recovery and removal

```
powershell.exe -NoProfile -ExecutionPolicy Bypass -File .\scripts\Repair-Axon.ps1
powershell.exe -NoProfile -ExecutionPolicy Bypass -File .\scripts\Uninstall-Axon.ps1
```

- Repair is idempotent and restores Synapse volume ownership without deleting data.
- Normal uninstall stops services and removes Axon firewall rules while preserving data.
- Permanent deletion requires -PurgeData plus the exact PURGE AXON confirmation and cannot be undone from Axon.

Control-panel-only update

Extract the Axon Control upgrade ZIP on the Windows host, double-click Update Axon Control.cmd, and approve the administrator prompt. The updater validates its payload, stops only the host-only control task, backs up the installed control panel under %ProgramData%\Axon\control-backups, installs the update, verifies it, and rolls back automatically if TCP 8780 does not return.